Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Statement-1: If a hyperbola does not have mutually perpendicular tangents, then its eccentricity is greater than $\sqrt{2}$.

Statement-2: Locus of a point from which two perpendicular tangents can be drawn to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2$.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.2

If the tangent at the point $P(a\sec\theta, b\tan\theta)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point where a directrix of the hyperbola meets the positive side of the transverse axis, then θ is equal to
Where e is the eccentricity of the hyperbola.

(A) $\cos^{-1}(1/e)$

(B) $\tan^{-1}(1/e)$

(C) $\cot^{-1}(1/e)$

(D) $\sec^{-1}(1/e)$

Q.3

If two perpendicular tangents can be drawn from the point (α, β) to the hyperbola $x^2 - y^2 = a^2$, then (α, β) lies on

(A) $y = \pm x$

(B) $x^2 + y^2 = a^2$

(C) $x^2 + y^2 = 2a^2$

(D) $y^2 = 4ax$

Q.4

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The tangent at an extremity (in the first quadrant) of latus rectum of the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$ meets x-axis and y-axis at A and B respectively. Then $(OA)^2 - (OB)^2$, where O is the origin, is equal to

(A) $-\frac{20}{9}$

(B) $\frac{16}{9}$

(C) 4

(D) $-\frac{4}{3}$

Q.5

The locus of the foot of the perpendicular drawn from the origin to any tangent to the hyperbola $\frac{x^2}{36} - \frac{y^2}{16} = 1$ is

(A) $\left(x^2 + y^2\right)^2 = 36x^2 - 16y^2$

(B) $\left(x^2 - y^2\right)^2 = 36x^2 - 16y^2$

(C) $\left(x^2 + y^2\right)^2 = 36x^2 + 16y^2$

(D) $\left(x^2 - y^2\right)^2 = 36x^2 + 16y^2$

Q.6

The locus of the middle points of the portions of the tangents of the hyperbola. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ included between the axes is

(A) $a^2x^2 - b^2y^2 = x^2y^2$

(B) $b^2x^2 - a^2y^2 = x^2y^2$

(C) $b^2x^2 - a^2y^2 + 4x^2y^2 = 0$

(D) $a^2x^2 - b^2y^2 + 4x^2y^2 = 0$

Q.7

If the slope of a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $2\sqrt{2}$, then the eccentricity e of the hyperbola lies in the interval.

(A) $(1, \sqrt{2}]$

(B) $(1, 2\sqrt{2}]$

(C) $(1, 3]$

(D) $(1, 4]$

Q.8

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If the tangent at the point $(a\sec\alpha, b\tan\alpha)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the transverse axis at T, then the distance of T from a focus of the hyperbola is

(A) $a(e - \cos\alpha)$

(B) $b(e + \cos\alpha)$

(C) $a(e + \cos\alpha)$

(D) $\sqrt{a^2e^2 + b^2\cot^2\alpha}$

Q.9

The line $2x + y = 1$ touches a hyperbola and passes through the point of intersection of a directrix and the x -axis. The equation of the hyperbola is

(A) $\frac{x^2}{1} - \frac{y^2}{3} = 1$

(B) $\frac{x^2}{1} - \frac{y^2}{3} = 2$

(C) $\frac{x^2}{3} - \frac{y^2}{1} = 1$

(D) $\frac{x^2}{3} - \frac{y^2}{1} = 2$

Q.10

The locus of the point of intersection of any two perpendicular tangents to the hyperbola is a circle which is called the director circle of the hyperbola, then the eqn of this circle is

(A) $x^2 + y^2 = a^2 + b^2$

(B) $x^2 + y^2 = a^2 - b^2$

(C) $x^2 + y^2 = 2ab$

(D) None of these

Q.11

Tangents at any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cut the axes at A and B respectively. If the rectangle OAPB (where O is the origin) is completed, then locus of point P is given by:

(A) $\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$

(B) $\frac{\mathrm{a}^2}{\mathrm{x}^2} + \frac{\mathrm{b}^2}{\mathrm{y}^2} = 1$

(C) $\frac{a^2}{y^2} - \frac{b^2}{x^2} = 1$

(D) None of these

Q.12

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The line $3x + 2y + 1 = 0$ meets the hyperbola $4x^2 - y^2 = 4a^2$ in the points P and Q. The coordinates of the point of intersection of the tangents at P and Q are

(A) $\left(-3a^2, 8a^2\right)$

(B) $\left(3a^2, 8a^2\right)$

(C) $\left(3a^2, -8a^2\right)$

(D) None of these

Q.13

If the straight line $x\cos\alpha + y\sin\alpha = p$ be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then

(A) $a^2\cos^2\alpha + b^2\sin^2\alpha = p^2$

(B) $a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$

(C) $a^2\sin^2\alpha + b^2\cos^2\alpha = p^2$

(D) $a^2\sin^2\alpha - b^2\cos^2\alpha = p^2$

Q.14

If the line $y = mx + \sqrt{a^2m^2 - b^2}$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point φ. Then $\varphi =$

(A) $\sin^{-1}(\mathrm{m})$

(B) $\sin^{-1}\left(\frac{\mathrm{a}}{\mathrm{bm}}\right)$

(C) $\sin^{-1}\left(\frac{\mathrm{b}}{\mathrm{am}}\right)$

(D) $\sin^{-1}\left(\frac{\mathrm{bm}}{\mathrm{a}}\right)$

Q.15

The auxiliary equation of circle of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

(A) $x^2 + y^2 = a^2$

(B) $x^2 + y^2 = b^2$

(C) $x^2 + y^2 = a^2 + b^2$

(D) $x^2 + y^2 = a^2 - b^2$

Answers & Solutions

Q.1 Answer:

STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

Solution:

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Statement- 2 is true as $x^2+y^2=a^2-b^2$ is the director circle of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ using it in statement-1, if the hyperbola does not have mutually perpendicular tangents then the points on $x^2+y^2=a^2-b^2$ are not real. $\Rightarrow b^2>a^2 \Rightarrow e^2=\frac{a^2-b^2}{a^2}>2 \Rightarrow e>\sqrt{2}$ and thus statement-1 is also true.

Q.2 Answer:

$\cos^{-1}(1/e)$

Solution:

Equation of the tangent at $P(a\sec\theta, b\tan\theta)$ to the hyperbola is
$\frac{x}{a}\sec\theta-\frac{y}{b}\tan\theta=1$ **(1)**
which passes through the point of intersections of the directrix $x=\frac{a}{e}$ and $y=0$, the transverse axis of the hyperbola.
i.e. $\left(\frac{a}{e},0\right)$. So from (1) we have $\frac{a}{a\times e}\sec\theta=1 \Rightarrow \sec\theta=e \Rightarrow \cos\theta=\frac{1}{e}$
$\Rightarrow \theta=\cos^{-1}(1/e)$.

Q.3 Answer:

$y=\pm x$

Solution:

An equation of a tangent to the hyperbola $x^2-y^2=a^2$ is
$y=mx+a\sqrt{m^2-1}$ **(1)**
Equation of the tangent perpendicular to (1) is $y=-\frac{1}{m}x+a\sqrt{\frac{1}{m^2}-1}$
$\Rightarrow my+x=a\sqrt{1-m^2}$ **(2)**
We must have $m^2-1\geq 0$ and $1-m^2\geq 0 \Rightarrow m^2=1 \Rightarrow m=\pm 1$ and thus (α,β) lies on $y=\pm x$.

Q.4 Answer:

$-\frac{20}{9}$

Solution:

Coordinates of the extremity of latus rectum in the first quadrant is $P\left(ae,\frac{b^2}{a}\right)=\left(3,\frac{5}{2}\right)$.
Equation of the tangent at P is $\frac{3x}{4}-\frac{5}{2}\left(\frac{y}{5}\right)=1$ or $3x-2y=4$.
which meets x-axis at $A\left(\frac{4}{3},0\right)$ and y-axis at $B(0,-2)$.
So $(OA)^2-(OB)^2=\frac{16}{9}-4=-\frac{20}{9}$.

Q.5 Answer:

$\left(x^2+y^2\right)^2=36x^2-16y^2$

Solution:

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Any tangent to the hyperbola is

$y = mx + \sqrt{36m^2 - 16}$ **...(1)**

Equation of the line through the origin perpendicular to the tangent is

$y = -\frac{1}{m}x$. **...(2)**

Eliminating m from (1) and (2) we get the required locus as

$$y = -\frac{x^2}{y} + \sqrt{36\frac{x^2}{y^2} - 16}$$

$$\Rightarrow \quad \left(x^2 + y^2\right)^2 = 36x^2 - 16y^2$$

Q.6 Answer:

$b^2x^2 - a^2y^2 + 4x^2y^2 = 0$

Solution:

Equation of a tangent to the hyperbola is $\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$ **which meets the axes at** $A(a\cos\theta, 0)$ **and** $B(0, -b\cot\theta)$.

If (h, k) **is the midpoint of** AB **then** $h = \frac{a\cos\theta}{2}, k = \frac{-b\cot\theta}{2}$.

Eliminating θ**, we get the required locus**

$$\left(\frac{a}{2h}\right)^2 - \left(\frac{-b}{2k}\right)^2 = 1$$

$$\Rightarrow b^2x^2 - a^2y^2 + 4x^2y^2 = 0$$

Q.7 Answer:

$(1, 3]$

Solution:

Any tangent to the hyperbola is

$y = mx \pm \sqrt{a^2m^2 - b^2}$

we must have $a^2m^2 - b^2 \geq 0$.

$\Rightarrow m^2 \geq \frac{b^2}{a^2}$

$\Rightarrow \quad m^2 \geq e^2 - 1$

$\Rightarrow \quad 1 < e \leq \sqrt{1 + m^2}$

$\Rightarrow \quad 1 < e \leq 3$ **when** $m = 2\sqrt{2}$.

Q.8 Answer:

$a(e - \cos\alpha)$

Solution:

Equation of the tangent is $\frac{x}{a}\sec\alpha - \frac{y}{b}\tan\alpha = 1$

Coordinates of T **are** $(a\cos\alpha, 0)$ **and of a focus are** $(ae, 0)$.

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Distance of T from the focus $(ae - a\cos\alpha) = a(e - \cos\alpha)$
(Note $e > \cos\alpha$)

Q.9 Answer:

$\frac{x^2}{1} - \frac{y^2}{3} = 1$

Solution:

Let the equation of the hyperbola be $x^2/a^2 - y^2/b^2 = 1.$
As the line $2x + y = 1$ touches the hyperbola $(2)^2a^2 - b^2 = 1$
$\Rightarrow 4a^2 - b^2 = 1$
$\Rightarrow \quad 4a^2 + a^2\left(1 - e^2\right) = 1, e$ being the eccentricity.
$\Rightarrow 5a^2 - a^2e^2 = 1$
As the line passes through $(a/e, 0).$
$2ale + 0 = 1 \Rightarrow a = e/2.$
So that $5\frac{e^2}{4} - \frac{e^4}{4} = 1$
$\Rightarrow e^4 - 5e^2 + 4 = 0$
$\Rightarrow \quad e^2 = 1$ or $e^2 = 4. \Rightarrow e = 2$ as $e^2 \neq 1$
Hence $a = 1$ and $b^2 = 3$ and the required equation is
$\frac{x^2}{1} - \frac{y^2}{3} = 1$

Q.10 Answer:

$x^2 + y^2 = a^2 - b^2$

Solution:

Equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ Any tangent to hyperbola are $y = mx \pm \sqrt{a^2m^2 - b^2}$ Also tangent perpendicular to this is $y = \frac{-1}{m}x \pm \sqrt{\frac{a^2}{m^2} - b^2}$
Eliminating m, we get $x^2 + y^2 = a^2 - b^2$

Q.11 Answer:

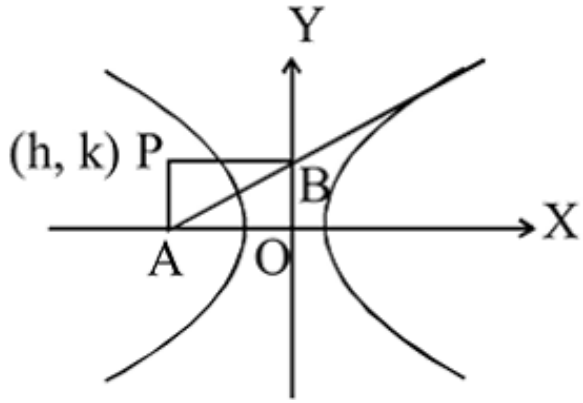
$\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$

Solution:

Eq. of the tangent at the point ' θ ' is $\frac{\mathrm{x}\sec\theta}{\mathrm{a}} - \frac{\mathrm{y}\tan\theta}{\mathrm{b}} = 1$
$\Rightarrow$ A is $(\mathrm{a}\cos\theta, 0)$ and B is $(0, -\mathrm{b}\cot\theta)$
Let P be $(\mathrm{h}, \mathrm{k}) \Rightarrow \mathrm{h} = \mathrm{a}\cos\theta, \mathrm{k} = -\mathrm{b}\cot\theta$
$\Rightarrow \frac{\mathrm{k}}{\mathrm{h}} = -\frac{\mathrm{b}}{\mathrm{a}\sin\theta} \Rightarrow \sin\theta = -\frac{\mathrm{bh}}{\mathrm{ak}}$ and

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$\cos\theta = \frac{h}{a}$



Square and add, $\Rightarrow \frac{b^2h^2}{a^2k^2} + \frac{h^2}{a^2} = 1 \Rightarrow \frac{b^2}{k^2} + 1 = \frac{a^2}{h^2}$
$\Rightarrow \frac{a^2}{h^2} - \frac{b^2}{k^2} = 1$
Hence, locus of P **is** $\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$

Q.12 Answer:

$(-3a^2, 8a^2)$

Solution:

Let the required point be (x_1, y_1)**. The given line** $3x + 2y + 1 = 0$ **...(1)**
is chord of contact of the point so it must be same as the line $T = 0$**, i.e.** $4xx_1 - yy_h = 4a^2$ **...(2)**
Comparing the coefficients of (1) **and** (2)**, we get**
$\frac{4x_1}{3} = \frac{-y_1}{2} = \frac{4a^2}{-1} \Rightarrow x_1 = -3a^2, y_1 = 8a^2$

Q.13 Answer:

$a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$

Solution:

$x\cos\alpha + y\sin\alpha = p \Rightarrow y = -\cot\alpha.x + p\,\text{cosec}\,\alpha$
It is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **Therefore,** $p^2\,\text{cosec}^2\alpha = a^2\cot^2\alpha - b^2 \Rightarrow$
$a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$

Q.14 Answer:

$\sin^{-1}\left(\frac{b}{am}\right)$

Solution:

(c) Equation of tangent at point ' φ **' on the**
hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **is**
$\frac{x}{a}\sec\varphi - \frac{y}{b}\tan\varphi = 1$ **or**
$y = \frac{b}{a}x\,\text{cosec}\,\varphi - b\cot\varphi$ **...(1)**
If $y = mx + \sqrt{a^2m^2 - b^2}$ **...(2)**
also touches the hyperbola then on comparing

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(1) \& (2)

$1 = \frac{\frac{b}{a}\,\mathrm{cosec}\,\varphi}{m} = \frac{-b\cot\varphi}{\sqrt{a^2\,m^2 - b^2}}$

Hence, $m = \frac{b}{a}\,\mathrm{cosec}\,\varphi$; **or** $\mathrm{cosec}\,\varphi = \frac{am}{b}$

or $\sin\varphi = \frac{b}{am}$, **or** $\varphi = \sin^{-1}\frac{b}{am}$

Q.15 Answer:

$x^2 + y^2 = a^2$

Solution:

The equation is $(x-0)^2 + (y-0)^2 = a^2$.

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